FTEC5660 REPRODUCIBILITY WORK

# A-MEM Memory Evolution Ablation

Testing full A-MEM against a link-only variant on LoCoMo

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# The paper claims that Memory Evolution improves A-MEM

**Multi-Hop F1 from the paper**

**27.02**
Full A-MEM

**21.35**
w/o ME

**3 seeds**
tested locally

## A-MEM pipeline

01 **Note** — 02 **Retrieve** — 03 **Link** — 04 **Evolve**

Scope: official benchmark code, official LoCoMo data, DeepSeek Chat, one conversation, 25 questions.

# One isolated change: remove neighbor updates

## FULL A-MEM

1. Builds structured memory notes
2. Retrieves similar memories
3. Creates links between related notes
4. Updates neighbor context and tags

## LINK-ONLY / w/o ME

1. Uses the same note construction
2. Uses the same retrieval and link step
3. Disables neighbor context updates
4. Disables neighbor tag updates

**The ablation is isolated and measurable: only Memory Evolution changes.**

## Overall mean F1

**41.76**
Full A-MEM

**38.25**
Link-only

**+3.51**
F1 points

## Paper vs this run

| Target metric | Paper GPT-4o-mini | This run DeepSeek |
|---|---|---|
| **Multi-Hop full** | 27.02 | 44.24 |
| **Multi-Hop w/o ME** | 21.35 | 28.69 |
| **Temporal full** | 45.85 | 11.59 |
| **Temporal w/o ME** | 31.24 | 12.62 |

**Multi-Hop direction reproduced. Temporal did not.**

# The pipeline worked. The paper's broad advantage did not.

## WORKED

- Six runs completed
- 25 questions per run
- Zero LLM failures
- Multi-Hop mean gain +15.56

## DID NOT HOLD

- Temporal mean change -1.03
- Single-Hop mean change -11.43
- Large seed-to-seed variance
- Small sample makes results uncertain

**Limits** DeepSeek, k=10, one conversation, 25 questions

**Conclusion: partial behavioral reproduction, not an exact numerical replication.**